Large Scale Synthetic Iris Generation for Few-Shot Recognition

MINOR PROJECT REPORT

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CERTIFICATE

This is to certify that the work embodied in this report entitled **"Large Scale Synthetic Iris Generation for Few-Shot Recognition"** has been satisfactorily completed by Ashish Yadav (Scholar no. 24U020044) & Divyam Agrawal (Scholar no. 24U020045). It is a bonafide piece of work, carried out under our guidance in the **Department of Computer Science and Engineering**, **Indian Institute of Information Technology, Bhopal** for the partial fulfillment of the Bachelor of Engineering during the academic year 2026-27.

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DECLARATION

We hereby declare that the following major project synopsis entitled “Large Scale Synthetic Iris Generation for Few-Shot Recognition” presented in the is the partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering**. It is an authentic documentation of our original work carried out under the able guidance of **Dr. Meenakshi Choudhary**. The work has been carried out entirely at the Indian Institute of Information Technology, Bhopal. The project work presented has not been submitted in part or whole to award of any degree or professional diploma in any other institute or organization.

We, with this, declare that the facts mentioned above are true to the best of our knowledge. In case of any unlikely discrepancy that may occur, we will be the ones to take responsibility.

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AREA OF WORK

Our project primarily focuses on **Biometric Recognition, Computer Vision, and Machine Learning**, with an emphasis on generating synthetic iris images for improving iris recognition under few-shot learning conditions. The work involves the use of image processing techniques, deep learning architectures, and generative models to develop a framework for synthetic iris generation and recognition. The project combines various image-processing and deep-learning techniques to analyze iris images, extract meaningful features and generate additional synthetic samples. It involves iris segmentation, normalization, image enhancement and quality assessment to prepare the datasets for subsequent experiments. Variational Autoencoders and conditional diffusion models are explored for learning identity-related representations and generating synthetic iris images. The work also focuses on evaluating the generated images in terms of identity consistency, visual quality and diversity. Different recognition models are trained and compared using real images, conventional data augmentation and synthetic-data-assisted training. Cross-dataset evaluation is additionally performed to examine the generalization of the developed models.

Overall, the project lies at the intersection of **Computer Vision, Generative Artificial Intelligence, Deep Learning and Biometric Security**. It aims to investigate the practical usefulness of synthetic data in improving iris recognition under few-shot learning conditions.

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Abstract

Iris recognition is a key form of biometrics for identifying individuals using their unique patterns found in their irises. However, creating a successful iris recognition system can be difficult when limited data is available. There are various challenges associated with collecting large biometric data sets to train systems, such as cost to collect, privacy issues and varying capture conditions. This project looks into using synthetic iris data to overcome limited data.

The aim of this project is to investigate a synthetic data assisted few shot iris recognition system to see if extra samples create better recognition results. This will involve: exploring iris data sets, preprocessing iris images, creating few shot learning scenarios, generating synthetic iris data, evaluating samples and training and evaluating recognition models. Experiments will be conducted using CASIA-Iris-Thousand and IIT Delhi iris datasets. Models will be compared using real only data, traditional augmentation and synthetic data assisted training. Cross data set experiments will be carried out.

Various experiments will be carried out to see the effect of synthetic data. These include comparison experiments, ablation studies and error analysis. This will allow us to understand when synthetic data can be useful and if there are any limitations. The aim of this project is to look into a practical solution to overcome issues with limited data. It will focus on creating a reliable iris recognition system using limited real samples per identity.

Introduction

1.1 Background

Iris recognition is a biometric method of identifying or verifying individuals based on the patterns present within their irises. The iris consists of extremely detailed texture making it an excellent feature to determine the difference between individuals. Iris recognition is a completely different approach as opposed to approaches looking at the appearance of an individual. This is a significant research area within computer vision and biometric security. An iris recognition system will typically consist of multiple steps when processing the iris. First the iris must be localised. Then unwanted areas of the image will be removed. After the iris has been extracted it will need to be normalized. Finally certain features will be identified to allow recognition. The quality of the output will depend on how good the image is and how well processed and recognised it has been [1].

Recently with the rise of deep learning neural networks have become a popular choice for iris recognition. These networks extract features from the images to allow the model to identify different iris identities. However, creating a good iris recognition model is not an easy task. The iris image can suffer many issues such as reflections, occlusion, varying lighting and other image acquisition issues. These can cause issues with the quality of features extracted and therefore affect the difficulty of the recognition. Recently there has been work in applying deep learning to iris segmentation and recognition. There has also been work looking at issues within the data such as quality, robustness and generalisation [2].

1.2 Research Problem and Motivation

A key issue faced when developing a strong iris recognition system is the large amount of training images per individual. Overall, deep learning systems require high-quality training data that will enable them to learn in a useful manner. If there is not enough training data, then the variations present in real images may not be captured by the model and it can become susceptible to overfitting. Unfortunately, this will affect the model's performance with unseen data. This problem forms the basis of a ‘few shot’ recognition problem. In this problem we are aiming to improve the recognitions system based on only a small amount of labelled examples. There has been past work completed into ‘few shots’ learning performed for iris; however, studies have shown that a lack of training data is a major issue stem from this issue [3].

Collecting and generating further biometric images can become problematic due to organisation; ensuring they matched to the correct identities and ensuring they are of the correct type for recognition. The quality of the samples also need to be considered, further complicating the matter. Improving the amount of training data is not always effective due to these factors. The additional data provided should still contain meaningful differentiation between different identities.

Creating synthetic images can be another method to overcome this constraint. We can create some additional samples to complete with training the model by using artificial samples. This will introduce some diversity to the training images, allowing the model to learn more robust representations. Work has been done into synthetical causing with iris images. For the generation, the quality, textures and recognition should be put in place. Many samples generated may look realistic to use but are not of any use to a recognition system. When creating synthetic samples, it is very important to create some meaningful differentiation between them, whilst ensuring they contain the features that allow them so be identified. Ultimately, their use should be determined through experimentation with recognition [4].

1.3 Overview of the Project

This project looks at the effectiveness of using synthetically generated iris images when we have very few real-world training examples per identity. The aim is to see if by including the synthetically generated examples into the training set, we can improve the recognition performance and understand how well the models perform under different evaluation scenarios.

The CASIA-Iris-Thousand and IIT Delhi iris datasets are used. The project begins with exploring the data and carrying out some pre-processing of the images. This includes segmenting the iris, normalizing, enhancing, and determining the quality of the iris images. Once the pre-processing has been completed, the data are then used to create several few-shot recognition scenarios and create a baseline. Synthetic examples are created, and the visual quality, variety, and suitability for recognizing identities is assessed.

The evaluation section looks at how the models perform under different training scenarios. This includes scenarios that are trained using only real images, standard data augmentation, and synthetic data assisted training. Experiments were conducted across different datasets to see how useful the learned representations were when evaluating on images from a different dataset. Comparisons were made, ablation studies were carried out and recognition performance was analysed along with failure cases. This will provide an understanding of how beneficial and limitations there are to using synthetic data in the few shot iris recognition scenario.

1.4 Objective and Scope

The purpose of this project is to investigate how synthetic iris can be used to overcome the issue of having very little training samples for iris recognition. This will be done by creating iris datasets, developing a synthetic data generation pipeline and analyzing how the samples perform.

Specifically:

1. Prepare the iris images from the chosen datasets for iris recognition.
2. Determine baseline performance when there is a limited amount of training samples.
3. Generate synthetic iris samples to analyze how visually appealing, varied and recognizable they are.
4. Analyze how well iris recognition performs using real training data, traditional data augmentation and synthetic data aided training.
5. Analyze how well the model generalizes using cross dataset experiments and investigate any limitations in the approach.

The scope of this project is to develop and investigate the complete workflow for generating synthetic data for iris recognition. It will not assume that the synthetic data will improve recognition performance. Instead, it will determine how useful the samples are in this particular situation. It will also determine where more work can be done.

Literature Review

2.1 Introduction

Iris recognition is a popular area of study within the field of biometrics due to the distinctiveness of the textural features present in the iris. Initial research focused on developing methods for isolating the iris region, describing the textured information and comparing different iris images. With an increase in computational power, recent research has looked to develop techniques to automatically extract features, improve recognition in varying imaging conditions and generate synthetic images for training biometric models [1].

This review relates to this projects research in four main areas: standard iris recognition, deep learning and few shot learning, synthetic iris image generation and evaluating the generated biometric data. All sections are very closely related. The type of recognition method will define the features required in the image, few shot learning deals with having very limited examples, while synthetic image generation can be used to increase the variety of the training data. Synthetic data can only be useful if it improves learning and does not create any deceptive patterns.

The following review provides an overview of the key developments within these areas and highlights the problem this project is trying to solve.

2.2 Conventional Iris Recognition Techniques

One of the key contributions to iris recognition was that of Wildes [5] who looked at the application of iris texture for automated personal identification and verification. His work described a recognition system comprising image acquisition, iris localisation, texture representation and then comparison of the images. It was his work which established the importance of good image processing prior to any attempt at person identification from an iris image.

Daugman [1] came up with a very well-known approach which was to encode the iris texture using multiscale quadrature wavelets. The resulting iris code is a representation of the distinctive information derived from the iris region and matching is achieved by comparing the codes. This work showed how the iris texture could be represented in a very compact form which was ideal for biometric matching. It also laid the groundwork for a great deal of the subsequent work in the areas of iris localisation, feature extraction and matching.

Ma et al. [6] looked at iris recognition via texture analysis and used spatial filters to extract the discriminative information from the iris images. They also took image quality into account when selecting the images to be used for recognition. This is of course pertinent as an image may contain an iris pattern that is clearly discernible but still be of poor enough quality to be

unusable for reliable matching (e.g. it may be blurred or otherwise affected by the acquisition process). Masek [7] also described an implementable iris recognition system which involved segmentation, normalization, feature extraction and matching and is therefore a useful reference point for the development of a conventional iris processing pipeline.

A fairly common step in these systems is iris normalization. The iris is roughly circular but its visible size varies due to pupil dilation, acquisition conditions and image position. Normalization maps the segmented iris region into a standardised form so that the features can be compared more consistently. Segmentation identifies the inner and outer boundary of the iris and quality assessment is used to flag images in which the relevant texture is obscured or otherwise of poor quality [1].

Although the conventional methods have established quite sound principles for iris recognition, their performance is still susceptible to segmentation errors, image quality and variations in acquisition conditions. This has led to a number of researchers looking at learned feature representations and more flexible approaches to the processing of iris images. In this project segmentation, normalization, enhancement and quality assessment are all important as errors at any of these stages will impact both the training data that is generated and the recognition model that is subsequently trained on it.

2.3 Deep Learning for Iris Recognition

The field of deep learning has revolutionised how we extract features useful for recognition. Traditionally, feature extraction would involve using hand crafted filters/texture descriptors. However, the use of neural networks allows us to learn these features from the training data. These features can be very complex and can be hard to describe in terms of any chosen image features.

Nguyen, Proença and Alonso-Fernandez have produced a very thorough survey of deep learning for iris recognition. They look at neural network-based approaches to both iris segmentation and recognition and go on to discuss the various issues of robustness, presentation attacks and so on in the context of other biometric applications. The survey is a good summary of the breadth of work that has been done and the continuing relevance of image quality, model design and evaluation conditions [2].

Ma et al. and Kumar and Passi both demonstrated that the choice of feature extraction and matching technique can have an impact on recognition performance. Kumar and Passi looked at a number of different iris matching techniques and examined the effect of combining the results of those techniques to see if this would improve the authentication performance. Their work again shows that performance is not just a function of the input data, it's also a function of how the features are matched and how the matching results are combined [8].

Although significant progress has been made in deep learning, obtaining good training data can be very important. A model trained on a small set of images can learn very effective features for the training distribution which may no longer be effective when the data is acquired under different circumstances. In addition, a model may achieve very good performance if there are a number of images available for each identity but may not perform as well when fewer images are available.

This is very relevant when evaluating an iris recognition system across different datasets. Different datasets will clearly have different resolutions, quality, distributions and capture circumstances. Just because a model performs well on one dataset doesn't mean it will perform well on another. Therefore, cross dataset evaluation can be an effective way to see whether the model is learning genuine iris characteristics or merely exploiting the peculiarities of the training data.

2.4 Few-Shot Learning and Limited Training Data

The idea behind few shot learning is that you do not have many examples of each label for training. Can we still learn a useful representation or decision function?

This is a big issue in iris recognition. For each identity, there will only be a limited number of images available that can be used for training. These will not necessarily represent all the variations that you will get in real scenarios. As a result, there may not be enough information to learn the invariant features of an identity and the variation between different images of that identity.

Lei et al. looked at few-shot iris recognition using model-agnostic meta-learning (MAML). They were looking at the issue of overfitting that can occur when you train a deep learning model on a small amount of data. The paper shows that few-shot learning is in fact a viable research direction for iris recognition rather than just a problem that crops up in the more general area of image classification [3].

Snell, Swersky and Zemel proposed Prototypical Networks which is a few-shot learning approach that classifies examples by comparing their representations to the class prototypes in a learned feature space. The work is a clear illustration of a more general principle in few-shot learning: that learning a good representation and measuring the similarity between samples is often more useful than simply having a conventional classifier trained on a large number of labelled examples [9].

However, even with few shot learning techniques there is an issue with using sufficient quality training data. If few real images have been used to train the model, then there may not be

enough variations learned to represent the identity correctly. To overcome this issue there could be more varied data created using data augmentation or by generating synthetic data.

Standard data augmentation includes methods to modify data such as flipping, cropping, and warping. Whilst this does provide more variation within the data it will still relate to the original image and not necessarily produce any new iris textures. Synthetic data, however, involves creating extra images based on the iris image distribution.

This must be tested to ensure it works. Increasing the amount of data in the training set doesn't always mean increasing its quality. If the generated data is very similar and doesn't look realistic or maintain the differences between individuals, this could be very unhelpful and in fact cause issues with the recognition model.

2.5Iris Datasets and Data Preparation

The availability and organization of benchmark datasets is of course a major consideration in iris recognition research. The CASIA-Iris-Thousand dataset contains 20,000 iris images from 1,000 subjects at 640 x 480 pixels and was created to support research with a fairly large number of identities [10]. It includes variations due to glasses and specular reflections so is useful for research that goes beyond the ideal of perfectly uniform images.
The IIT Delhi Iris Database is another well-established database for iris recognition research. The database page itself describes the database as being a collection of images taken in an indoor environment with variable image quality [11]. The version of the database described on that page contains 2,180 images from 218 subjects at 320 x 240 pixels. Kumar and Passi [9] also used the IIT Delhi iris database in their comparative study of various recognition techniques.

Care should be taken when specifying the dataset. One paper mentions using a dataset of 1120 samples from 224 users from IIT Delhi. However, on the database page it mentions there are 2180 samples from 218 subjects. These are different datasets. In order to have a fully reproducible project report you should specify exactly which dataset you are using. Give information on the source, which version, who is included and how many samples there are.

Using both CASIA-Iris-Thousand and IIT Delhi data gives a good base for experimenting with different recognition scenarios. Using more data means we can experiment with a higher number of identities and using IIT Delhi gives us another source of iris data. How much benefit this provides will depend on the training and testing protocol used.

Preprocessing is an important stage. Segmentation allows us to extract just the iris from the whole eye image. Normalisation gives us a more consistent representation of the images. Enhancement can help make the texture clearer. Quality assessment helps us identify any

images which may not provide good features. Errors introduced during preprocessing or inconsistencies between the datasets will affect the comparison between real and synthetic samples.

Preprocessing should be considered part of the recognition process. It also helps us determine if recognition issues are due to the image, the quality of the generated sample or due to the model's learnt representation.

2.6 Development of Synthetic Iris Image Generation

Early work on synthetic iris generation was aimed at replicating the key textural features of the iris via statistical modelling and image construction. Shah and Ross [12] proposed a method based on feature agglomeration in which a Markov random field model was used to generate the background texture, with the various features of the iris (radial furrows, concentric furrows, crypts and the collarette) then added on. The aim was to produce synthetic iris images which could be used to test recognition algorithms.

Such methods demonstrated that the structure of an iris could be modelled computationally. The problem with manually specifying the texture components is that this in turn restricts the range of variation that can be represented in the generated samples. Producing convincing images with sufficient diversity is still a real problem, especially when you're trying to generate synthetic images that represent a wide range of biometric characteristics.

Deep generative models took a much more flexible approach. Goodfellow et al. [13] proposed Generative Adversarial Networks (GANs) which train a generator to produce samples and a discriminator to tell the generated samples apart from the real training data. The generator is trained in an adversarial manner so that it learns to produce samples that become increasingly indistinguishable from the training distribution.

GAN based approaches have been looked at subsequently for the generation of synthetic biometric data. Yadav, Chen and Ross [14] proposed a method based on a relativistic average GAN for the generation of synthetic iris images and looked at its use in the context of presentation attack detection. They showed that synthetic iris images could be of use in training biometric security systems rather than just as a visual example.

Yadav and Ross subsequently proposed iWarpGAN [15] which separates the identity and style information to generate synthetic iris images with both inter-class and intra-class variation. The distinction is of course important in the generation of biometric data. Inter-class variation is the variation between identities; intra-class variation is the variation between the images of the same identity. A useful synthetic dataset has to represent both of these forms of variation.

Yadav and Ross looked at a range of GAN based iris synthesis techniques and discussed the pros and cons of these in terms of visual realism, uniqueness and biometric utility. Their review makes it clear that assessment of synthetic iris data should not be confined to how realistic the images look. The images also need to be sufficiently diverse and useful for the application in question (whether that be recognition or presentation attack detection) [16].

Kordas et al. [4] looked at synthetic iris generation using StyleGAN3 and compared the image generation in Cartesian and polar space. They looked at image quality and texture and used iris recognition software to determine whether the generated samples actually contained some useful, discernible texture. This is a very relevant study in that it looks at the generation representation and how that relates to the requirements of the downstream task (iris recognition).

Overall, these studies highlight the move from manually designed textural models to learned generative approaches. The studies show that several questions must be addressed when evaluating synthetic iris images: Are they realistic? Do they have individual identities? Are their variations meaningful? Does including them in the training set improve recognition performance?

2.7 Modern Generative Models and their relevance

Recent research into image generation has shifted away from the use of an adversarial learning methodology. Ho, Jain and Abbeel [17] have proposed the use of denoising diffusion probabilistic models. These have demonstrated exceptional results on a variety of image benchmarks. This was a huge contribution to the field of deep generative modelling as it shows that we can find a process to de-noise a representation and generate a sample similar to those in our training distribution.

Whilst the results from Ho et al., [17] demonstrate the capability of this type of generative model, we cannot conclude that they are effective for iris recognition. Biometric images have a variety of requirements. The texture must contain enough detail with the correct amount of variation to be useful to a recognition system.

When using generative models to create images for general purpose, we must be cautious. Whilst a method may create highly realistic images when applied to a certain domain, a completely different dataset, pre-processing and evaluation criteria may be required to apply it to a biometric application. It is therefore extremely important to evaluate the models within a specific domain.

In regards to this particular area, it is not enough to know which generative model will create realistic samples. We must determine whether these generated images provide us with any

additional useful information for recognition when there are only a small amount of samples per identity in the real training set.

2.8 Evaluation of Synthetic data and Biometric Utility

For the synthetic iris data, the image quality should be considered. Are there any clear problems with the images? For example, are there distorted edges, repeating patterns, unnatural textures or irregularities in the iris? This can help to identify samples which are clearly not going to be suitable for training.

The variety of the samples should also be considered. The generator may be producing lots of images but are they all the same? Just because there are many output images does not mean that there are many unique or useful samples. Within the biometric context it will be an issue of whether the generator is unintentionally replicating the characteristics of specific training identities.

Tinsley, Czajka and Flynn [18] looked at identity leakage in synthetic iris images. They found that there is a reason why biometric generation has to be treated with care in this respect as synthetic images can in some cases be very similar to the actual training images and thus may be of concern in terms of identity information being reproduced. Generation quality and privacy properties should be assessed rather than taken for granted.

Quantitative image metrics can also be of use in supplementing visual inspection. Heusel et al. [19] proposed the Fréchet Inception Distance (FID) in the context of generative model evaluation. The FID works by computing the distance between the feature distributions of real and generated images (the distance being measured in terms of the Fréchet distance). It's a useful metric for comparing the distributions of generated images under a given evaluation regime.

However, it is important to highlight that the FID score alone will not tell you whether an image has conserved any identity related information and improved recognition. While two distributions might be similar when looked at, they might not be the same from a biometric perspective. Similarly, while an image might be different to the mean training image it may still have valid and useful iris textures. As such you should look at your images, as well as FID, and the performance in recognition.

The most direct way to see how useful your synthetic data has been is to train models on the different data types and use the correct metrics for your task. This could include looking at the classification accuracy for identification tasks, receiving operating characteristic curves, equal error rates for verification tasks or rank based accuracy if your evaluation protocol requires the candidates to be ranked by the model.

You should always ensure you are using the same split of data and using the same evaluation protocol. If your synthetic images are used in training but not evaluation then the results can help you understand how useful they have been for the model. Testing across datasets can give you evidence of generalisation and ablation studies can help you understand why certain elements are contributing.

2.9Comparative Summary of Related work

Table 1: Summary of Related Literature

Reference	Main contribution	Relevance to this project
Daugman [1]	Established iris coding and matching using phase information.	Provides the foundation for iris recognition and matching.
Nguyen et al. [2]	Reviewed deep learning methods for iris segmentation and recognition.	Explains current learning-based approaches and their challenges.
Lei et al. [3]	Investigated MAML-based few-shot iris recognition.	Establishes the relevance of limited-sample recognition.
Shah and Ross [12]	Generated synthetic iris textures using statistical modelling.	Represents early attempts at synthetic iris generation.
Yadav et al. [14]	Investigated GAN-generated iris images for presentation attack detection.	Demonstrates a biometric security application of synthetic data.
Yadav and Ross [15]	Developed a method for generating iris images with identity and style variations.	Highlights the importance of identity diversity and intra-class variation.
Kordas et al. [4]	Compared synthetic iris generation in Cartesian and polar representations.	Connects image representation, quality assessment, and recognition utility.
Ho et al. [17]	Demonstrated high-quality image generation using diffusion probabilistic models.	Provides background on an alternative family of generative approaches.
Tinsley et al. [18]	Examined identity leakage in synthetic iris images.	Motivates checking whether generated samples reproduce training identities.
Heusel et al. [19]	Introduced FID as a	Supports image-distribution

	quantitative measure for generative model evaluation.	comparison alongside biometric evaluation.

The studies reviewed above address different parts of the problem. Conventional recognition research establishes how iris features can be extracted and compared, while few-shot learning examines how recognition can be performed with limited labelled examples. Synthetic iris generation research investigates ways to produce additional samples, and quality and identity-leakage studies show why the usefulness of generated images cannot be established by appearance alone.

2.10 Research Gap and Relevance to the project

The literature shows that this is not a new problem. Previous statistical methods, GAN-based methods and recent generative models have shown to create synthetic iris images. There has been research into using synthetic data for recognition and biometric security. This shows that there is not a lack of work in this area.

How reliable can generated samples be for recognition? The generator must consider the properties of the generated images, the diversity of the generated dataset, and how this data was used for training.

Generalization: Even if good results can be achieved on one dataset, this does not guarantee that the recognition system will work well on another dataset with different image properties. Testing the recognition system on different datasets can help determine whether the model has learned useful features.

It may not be useful to generate a large number of images. The synthetic images may not provide any useful diversity to the dataset. The images may also contain artifacts and similarities between identities which can impact recognition. The properties of the images and performance of the recognition model must be considered.

This project looks into this by using a pipeline that preprocesses the iris images, performs baseline recognition, generates synthetic samples, verifies based on identity, determines diversity and compares recognition. Using CASIA-Iris-Thousand and IIT Delhi data allows us to see how the recognition performs using different data. Comparing only using real data and traditional augmentation will help determine if using synthetic samples provides any benefit.

This project looks into how useful synthetic iris data can be for few-shot recognitions.

Problem Definition and Objectives

3.1 Problem definition

Systems using iris recognition use the unique patterns found in the iris to identify/verify people. The quality and variety of training data used to train the system to extract discriminative features determines how well the system performs. In real world conditions there may be limited amounts of labelled iris data available to train the model for each identity. Training a model with limited examples could mean the system does not learn enough information to recognise a person when given a different image of their iris.

Few-shot learning has been used to try to improve this. Several methods have been tried to improve few shot iris recognition however having varied examples to train the model can be an issue. Data augmentation can help increase the number of examples by applying changes to the current images. However, these changes might not create enough extra texture variation. Another way to improve the amount of examples would be to generate synthetic iris images. However, these must be meaningful and contain variation to be effective.

The key issue to be explored in this project is whether synthetic iris images can help improve the recognition performance of a system with limited real examples to train for each identity. This will be investigated by exploring whether improvements can be seen consistently across different evaluation scenarios. This means synthetic examples must be tested to see how useful they are.

This project will use the CASIA-Iris-Thousand and IIT Delhi iris datasets. A workflow was setup including pre-processing, baseline recognition, synthetic example generation, identity verification, assessing diversity and comparing results. This will give insight into how well the models perform with different data conditions.

The aim of this study is not to assume more examples will always improve recognition however to understand whether the synthetic examples are providing useful extra training information. It will also highlight under what conditions they could be beneficial and what issues there are.

3.2 Objectives

Project Objective:

To study how well synthetic iris image generation can be used to improve the performance of few-shot iris recognition systems.

Specific Objectives:

1. Data Preparation: Explore and prepare iris training data from the CASIA-Iris-Thousand and IIT Delhi iris datasets.

2. Image Preprocessing: Implement necessary iris image preprocessing steps, such as segmentation, normalization, enhancement and quality analysis.

3. Baseline Recognition: Create a baseline recognition system and test its performance on a few-shot scenario.

4. Synthetic Data Generation: Create and implement a synthetic iris generation pipeline.

5. Sample Quality Analysis: Analyze the quality, diversity and identity characteristics of the generated samples.

6. Recognition Evaluation: Compare the performance of a recognition system trained using only real training data, traditional data augmentation, and synthetic data.

7. Generalization Evaluation: Evaluate the generalization capabilities of the model by performing cross dataset evaluation (where possible).

8. Experimental Analysis: Analyse how well the model performs using different experiments, ablation studies and failure analysis. This will help to determine what factors affect the performance of the model.

3.3 Scope of the Project

This project will cover producing iris datasets, preprocessing iris images, generating synthetic examples, and analysing these examples with recognition experiments. Image quality and diversity will be analysed, and compared with a base training set. How well the synthetic examples work across different datasets will also be investigated.

This project is focused on how useful synthetic data can be for few-shot iris recognition. The conclusions will be drawn from the experiments carried out. Any claims that synthetic data improve recognition accuracy, or can be created at scale, or protect privacy will have to be proven by experimental results.

Proposed Methodology and Work Description

4.1 Overview of the Proposed Methodology

A synthetic iris generation system for few shot iris recognition will be developed.

The system will be broken down into several components: pre-processing, disentangling, synthetic generation, anatomical constraints, expansion and finally few shot iris recognition.

Pre-processing is carried out on real iris images from CASIA-Iris-Thousand and IIT Delhi iris datasets. Images will be segmented, normalized and enhanced. Quality assessments will be done. Representations will be learnt which captures identity-related features whilst separating out any illumination, pupil size, capture device and image quality variations.

These representations will be used to condition a generative model. A new component will be introduced to determine if anatomical constraints have been maintained during image synthesis. The synthetic dataset will be used to train and test few shot recognition models. Finally, the performance of the recognition models will be analysed, alongside their generalisation capabilities and the limitations of using synthetic data to assist training.

4.2 Dataset Collection and Exploration

The first stage involves obtaining and examining the CASIA-Iris-Thousand and IIT Delhi iris datasets. Dataset exploration includes examining image dimensions, file formats, the distribution of images across identities, and the availability of usable samples. The initial analysis also considers image-quality variations and potential issues such as unreadable files, inconsistent image formats, and unsuitable samples.

4.3 Iris Image Preprocessing

Preprocessing is performed to prepare the iris images for recognition and synthetic-data experiments. It consists of the following stages.

4.3.1 Iris Segmentation

Iris segmentation aims to identify the iris region in an eye image while excluding irrelevant regions. First, pupil and limbus boundaries are detected, and then masks are constructed for eyelid occlusions, eyelash-like structures and specular reflections.

4.3.2 Iris Normalization

After segmentation, the iris region is converted into a standardized rectangular representation using Daugman's Rubber Sheet Model. The transformation maps points between the pupil and

limbus boundaries into normalized radial and angular coordinates. This process reduces geometric variation associated with differences in pupil size and iris boundary dimensions [1].

4.3.3 Image Enhancement

Image enhancement is applied to improve the visibility of relevant iris texture where necessary. It can help address certain image-quality issues, although it cannot recover all information lost through severe blur, occlusion, or poor acquisition.

4.3.4 Quality Assessment

The processed images are examined to identify samples that may be unsuitable for reliable analysis. Quality assessment helps identify potential problems such as insufficient texture visibility, blur, or processing artifacts. The quality-assessment procedure is used to understand the condition of the available images and to support consistent dataset preparation.

4.4 Construction of Few-Shot Recognition Scenarios

After preprocessing, the data is organized into training and evaluation sets. The central requirement is to simulate conditions in which only a small number of real training images are available for each identity.

Few-shot scenarios are constructed by restricting the number of real images used for training while retaining a separate set of images for evaluation [9]. This enables the study to examine recognition performance under limited-data conditions.

Careful consideration should be taken when splitting the data into train and test subsets. If there are multiple images of the same identity, then the train/test split should be performed according to the desired recognition protocol.

This provides a fair comparison between the different methods (pure-real, traditional augmentation, and synthetic data assisted).

4.5 Baseline Recognition

We create a baseline by training our recognition system with only the available real training data, that will act as a baseline to compare different training scenarios. This baseline is trained and tested using the chosen few shot protocol. We gather appropriate recognition metrics for the recognition task we have carried out during the experiments. This gives us an insight into how well the model will perform when training on limited real examples.

Creating this baseline is important as any improvements or worsenings in performance need to be compared to a clear reference point.

4.6Synthetic Image Generation

The next stage involves generating additional iris images for use in the recognition experiments. The purpose of synthetic generation is to investigate whether additional samples can provide useful training information when real images are scarce.

The generation workflow uses the prepared project data and the implemented generation procedure to produce synthetic iris samples. The resulting images are organized separately from the original datasets so that their origin can be identified and their contribution can be evaluated independently.

The generated samples are examined for obvious defects, repetitive patterns, and inconsistencies in texture. Their suitability is not determined solely by visual appearance. Instead, their usefulness is assessed through subsequent diversity analysis and recognition experiments.

The generated images are then incorporated into the relevant experimental configurations according to the training protocol. The methodology does not assume in advance that all generated images will be beneficial.

4.7Quality and Diversity Assessment of Synthetic Images

For synthetic image analysis we look at whether the data are appropriate for the desired recognition task. The main categories include:

- Quality analysis: Checking whether there are any artefacts in the samples. For example, distorted shapes, unnatural texture patterns etc.

- Variety analysis: Looking at whether there is enough variety in the generated images. It may be the case that although there are many samples, they are very similar.

- Identity analysis: If the experiments allow then identity verification will be used to determine if there are useful identity differences between the samples.

The results are based on both the quantitative analysis available and observing the samples. These can be used to understand how well the samples will perform in recognition, and what issues may be in the generation process.

4.8Comparative Recognition Experiments

A comparative experiment is performed using different training methods to determine how beneficial it is to train with synthetic data. Below are the 3 main methods used:

1. Training with only real data
Where the recognition model is trained on the chosen real training data.

2. Training with regular data augmentation
Where the model is trained on real data and with the applied regular data augmentation.

3. Training with synthetic data
Where the model is trained on real data and with a combination of synthetic data.

It is important to keep the evaluation consistent between the different training methods, so we can compare how they affect the results. The results will be compared using the correct metrics for the implemented recognition task. We will then look at if using synthetic data has improved the results, given similar results, or has any negative effects.

4.9 Cross-Dataset Evaluation

An analysis of cross-dataset performance will be conducted to determine if the recognition performance can be useful when the test images come from a different dataset. Caution has to be taken in interpreting these results as there could be various factors involved such as differences between the composition of the datasets, quality of the images and the parameters used during capture. This analysis provides further evidence of generalisation but does not prove that it will work for all biometric datasets.

4.10 Ablation Studies and Error Analysis

Ablation studies will be conducted to understand the contributions of specific components. This will be done by comparing different configurations of our experiments to understand which components contribute to changes in the recognition performance.

Error analysis will be conducted to understand where the recognition system failed to correctly predict. Where possible, we will analyse these cases based on the quality of the input images, pre-processing, properties of the synthetic samples and differences between data sets.

By using both the comparative results and error analysis, we can better understand the advantages and disadvantages of using synthetic data to assist recognition.

4.11 Overall Work Description

The work can be broken down into the following steps:

1. Collect and investigate the chosen iris datasets.

2. Prepare the data and perform preprocessing.

3. Create a few-shot learning scenario.

4. Establish a baseline for the iris recognition task using real training data.

5. Generate synthetic iris data.

6. Investigate the quality, variety and characteristics of the generated data.

7. Compare results from training on real data only, traditional data augmentation and training on synthetic data.

8. Perform cross dataset experiments.

9. Ablation studies and error analysis.

10. Conclude what was learnt from the experiments and what limitations there are, and potential improvements for the future.

This provides a clear structure for investigating synthetic data in few shot iris recognitions. The main goal of this is to see if the synthetic images provide useful information to the recognition model when there is less data available. This accounts for the quality, variety, evaluation and generalisation of the images.

Proposed Algorithms

5.1 Phase-1: Image Preprocessing

Table 2: Algorithms used in Phase-1

Task	Algorithm / Method Used	Fallback (if any)
Pupil boundary detection	Circular Hough Transform [28]	Thresholding and Connected Component Analysis
Limbus boundary detection	Hough Circle Transform with geometric constraints and radial gradient scoring	No alternative specified
Eyelid detection	Canny Edge Detection + Probabilistic Hough Line Transform	No alternative specified
Eyelash detection	Morphological Black-Hat Transform with horizontal and vertical kernels	No alternative specified
Reflection detection	Intensity thresholding (99.5th percentile, minimum 240) + Connected Component Analysis	No alternative specified
Final iris mask generation	Circular masking and Boolean mask operations	No alternative specified
Iris normalization	Daugman's Rubber Sheet Model [1]	No alternative specified
Global enhancement	Histogram Equalization	CLAHE
Local enhancement	CLAHE (8 × 8 grid, clip limit 2.0)	Histogram Equalization
Brightness adjustment	Gamma Correction ($\gamma = 1.2$)	No alternative specified
Contrast enhancement	Percentile-based Contrast Stretching (2nd–98th percentile)	Original image retained if upper percentile ≤ lower percentile
Blur assessment	Variance of Laplacian	No alternative specified
Focus assessment	Tenengrad Measure (Sobel Gradients)	No alternative specified
Contrast assessment	Standard Deviation of Valid-Pixel Intensities	No alternative specified
Illumination assessment	Mean Intensity of Valid Pixels	No alternative specified
Occlusion assessment	Invalid-Pixel Fraction	No alternative specified
Reflection assessment	Reflection-Area Fraction	No alternative specified

5.2 Phase-2: Few-shot Dataset Construction and Baseline Recognition

Table 3: Algorithms used in Phase-2

Task	Algorithm / Method Used	Fallback (if any)
Dataset splitting	Identity-wise Training, Validation and Testing Split	No alternative specified
Few-shot sampling	K-shot Sampling per Identity	No alternative specified
Baseline feature extraction	CNN-based feature extraction	ResNet50 / EfficientNet / ViT (candidate architectures)
Baseline classification	ArcFace-based recognition objective [24]	No alternative specified
Baseline evaluation	Accuracy, ROC and EER	No alternative specified

5.3 Phase-3: Identity-Disentangled Representation Learning

Table 4: Algorithms used in Phase-3

Task	Algorithm / Method Used	Fallback (if any)
Latent representation learning	Variational Autoencoder (VAE) [23]	β-VAE
Identity representation learning	Identity Classification Loss	No alternative specified
Identity separation	Triplet Loss	No alternative specified
Latent regularization	KL-Divergence Loss	No alternative specified
Image reconstruction	Reconstruction Loss	No alternative specified

5.4 Phase-4: Synthetic Iris Generation and Anatomical Assessment

Table 5: Algorithms used in Phase-4

Task	Algorithm / Method Used	Fallback (if any)
Synthetic image generation	Conditional Diffusion Model [17]	StyleGAN3
Latent-space generation	Latent Diffusion	Conditional Diffusion
Identity conditioning	Identity Embedding	No alternative specified
Appearance conditioning	Illumination, Pupil Dilation, Sensor and Occlusion Variables	No alternative specified
Anatomical boundary assessment	Boundary Consistency Loss	No alternative specified
Texture assessment	Texture Consistency Loss	No alternative specified
Structural assessment	Structural Consistency Loss	No alternative specified

Anatomical validation	Differentiable Anatomical Discriminator	Constraint-based structural assessment

5.5Phase-5: Synthetic Dataset Expansion

Table 6: Algorithms used in Phase-5

Task	Algorithm / Method Used	Fallback (if any)
Synthetic sample generation	Conditional Sampling from Trained Generator	No alternative specified
Image degradation	Motion Blur, Specular Reflection and Compression Artifacts	No alternative specified
Sensor variation	Sensor-Conditioned Generation (NIR and Visible Light)	No alternative specified
Dataset integration	Real and Synthetic Dataset Combination	No alternative specified
Sample filtering	Quality and Anatomical Assessment	No alternative specified

5.6Phase-6: Few-Shot Recognition and Evaluation

Table 7: Algorithms used in Phase-6

Task	Algorithm / Method Used	Fallback (if any)
Recognition backbone	EfficientNet-B3	ConvNeXt-Tiny / Swin Transformer
Feature representation	256-dimensional Embedding	No alternative specified
Few-shot classification	Prototypical Networks [9]	Relation Networks
Meta-learning	MAML [3]	No alternative specified
Contrastive representation learning	SimCLR / Supervised Contrastive Learning	No alternative specified
Prototype computation	Mean of Support-Set Embeddings	No alternative specified
Identity matching	Euclidean Distance	Cosine Similarity
Cross-dataset evaluation	Source-to-Target Dataset Evaluation	Reverse-direction evaluation

5.7Phase-7: Explainability, Error Analysis and Statistical Evaluation

Table 8: Algorithms used in Phase-7

Task	Algorithm / Method Used	Fallback (if any)
Model explainability	Grad-CAM (if using a compatible CNN) [26]	No alternative specified
Recognition error analysis	False Prediction and Misclassification Analysis	No alternative specified
Ablation study	Component-wise Ablation Experiments	No alternative specified
Statistical comparison	Appropriate Statistical Hypothesis Tests	Descriptive Statistical Analysis
Inference	Preprocessing + Feature Extraction + Identity Matching	No alternative specified

Proposed Workflow

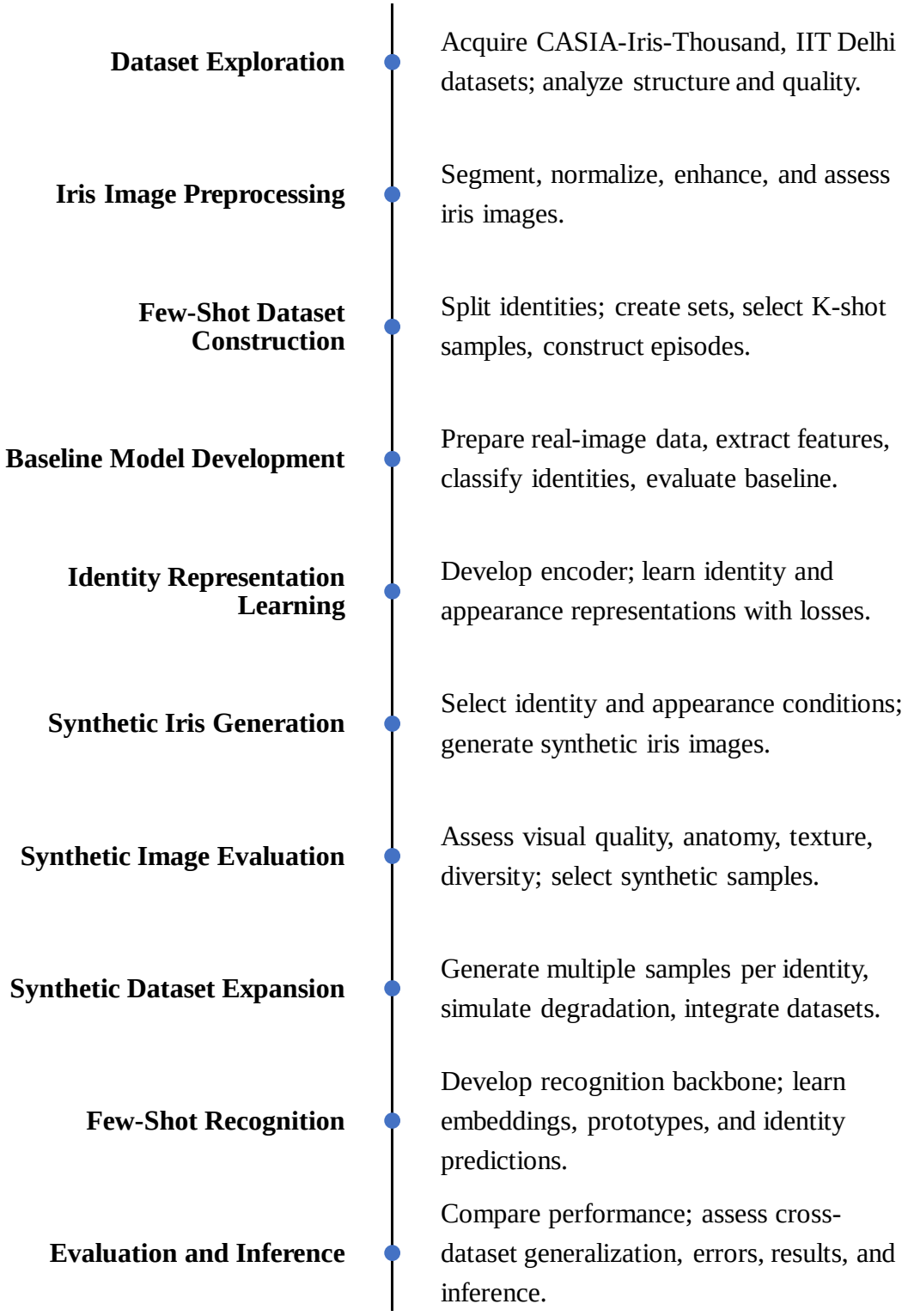


Figure 1: Workflow of the project

Implementation Details

[GitHub repository-Large Scale Synthetic Iris Generation for Few-Shot Recognition]

7.1 Development Environment

The project titled, Large-Scale Synthetic Iris Generation for Few-Shot Recognition, was written in Python, using Jupyter Notebook. It was created as multiple different notebooks so each part of the process can be clearly seen. This meant that the data processing, image processing, representation learning, synthetic image generation, recognition and analysis were all clearly separated.

The major libraries and technologies used are:

- Python: Main programming language.
- Jupyter Notebook: Development and experimentation.
- OpenCV: Image processing, segmentation and normalization.
- NumPy and Pandas: Numerical operations and dataset management.
- PyTorch and Torchvision: Deep-learning model development and training.
- Scikit-learn: Data processing and evaluation utilities.
- Matplotlib and Seaborn: Data visualization and result analysis.

7.2 Dataset Exploration – Notebook 01

Loaded and explored CASIA-Iris-Thousand and IIT Delhi dataset. Implemented methods to determine amount of individuals, images per individual, resolution of images, amount of left and right irises in the dataset and class imbalance.

Outputs: Dataset statistics, CSV reports, distribution plots and sample visualizations.

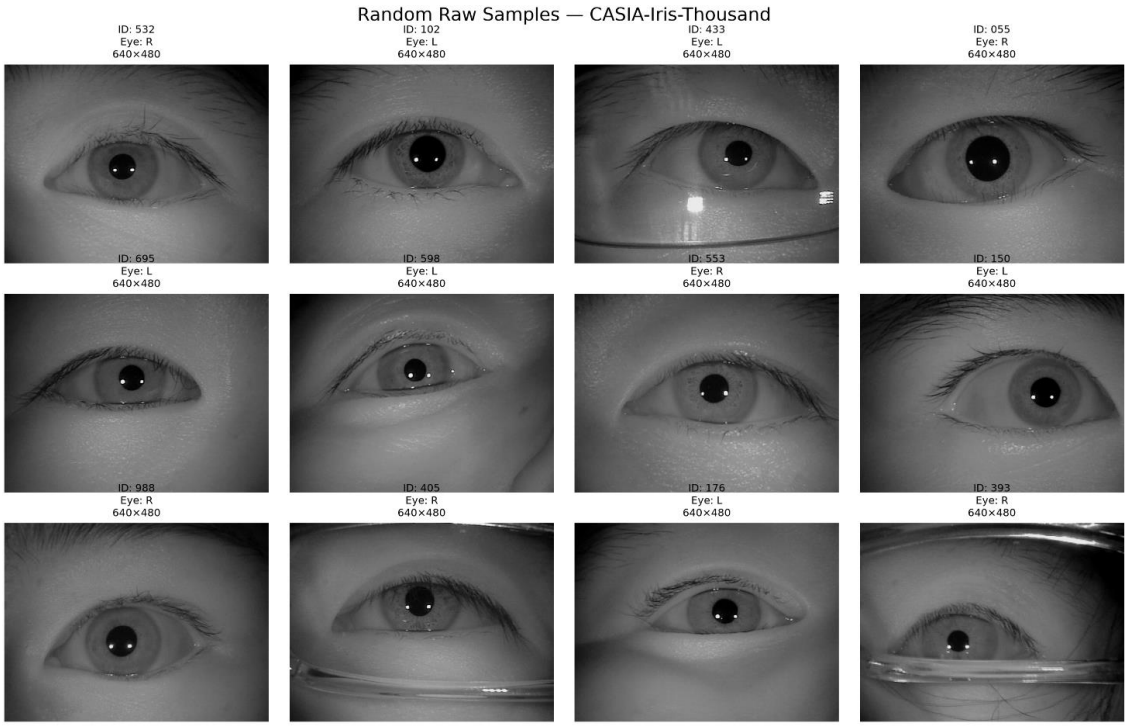


Figure 2: Samples of CASIA-Iris Thousand dataset

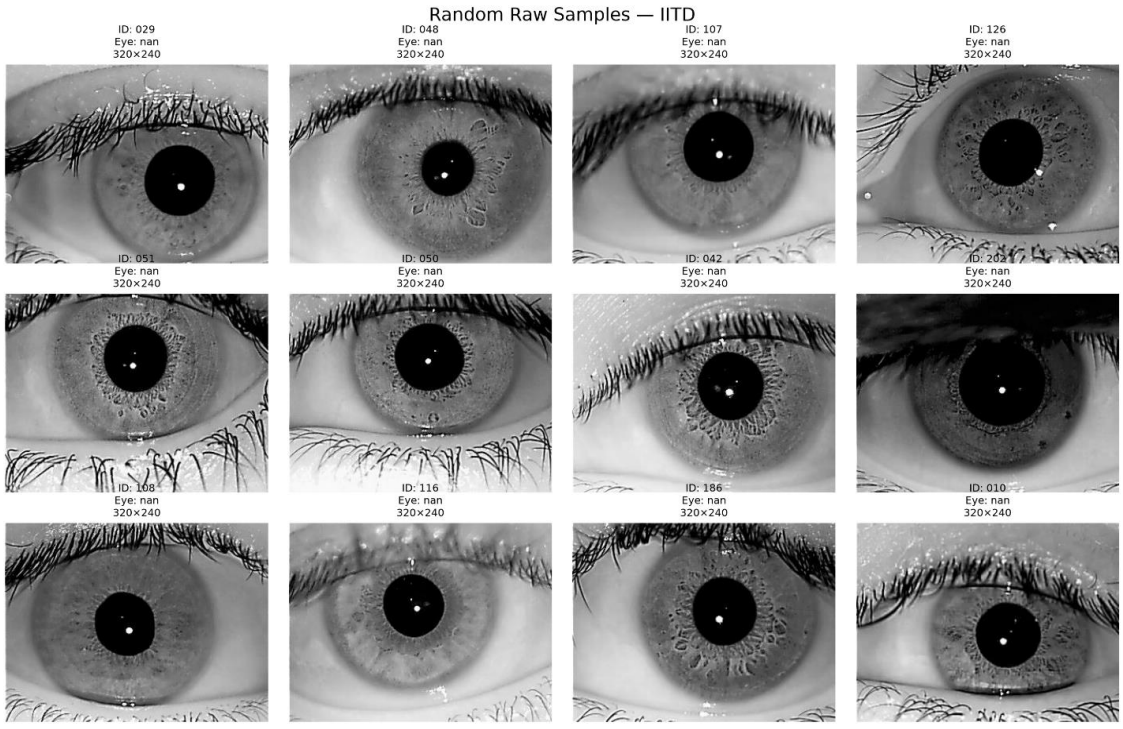


Figure 3: Samples of IIT Delhi Iris dataset

7.3 Iris Image Preprocessing

7.3.1 Iris Segmentation – Notebook 02

Iris segmentation was implemented using classical computer vision techniques in OpenCV.

- Pupil detection: Circular Hough Transform [28], with thresholding and connected-component analysis as fallback methods.
- Limbus detection: Hough Circle Transform with geometric constraints and radial gradient-based candidate selection.
- Eyelid masking: Gaussian blurring, Canny Edge Detection and Probabilistic Hough Line Transform.
- Eyelash masking: Morphological Black-Hat transformation, thresholding and connected-component filtering.
- Reflection masking: Intensity-based thresholding and connected-component analysis.

The detected unwanted regions were combined to construct a final mask containing the usable iris region.

Output: Segmented iris images, binary masks and segmentation metadata.

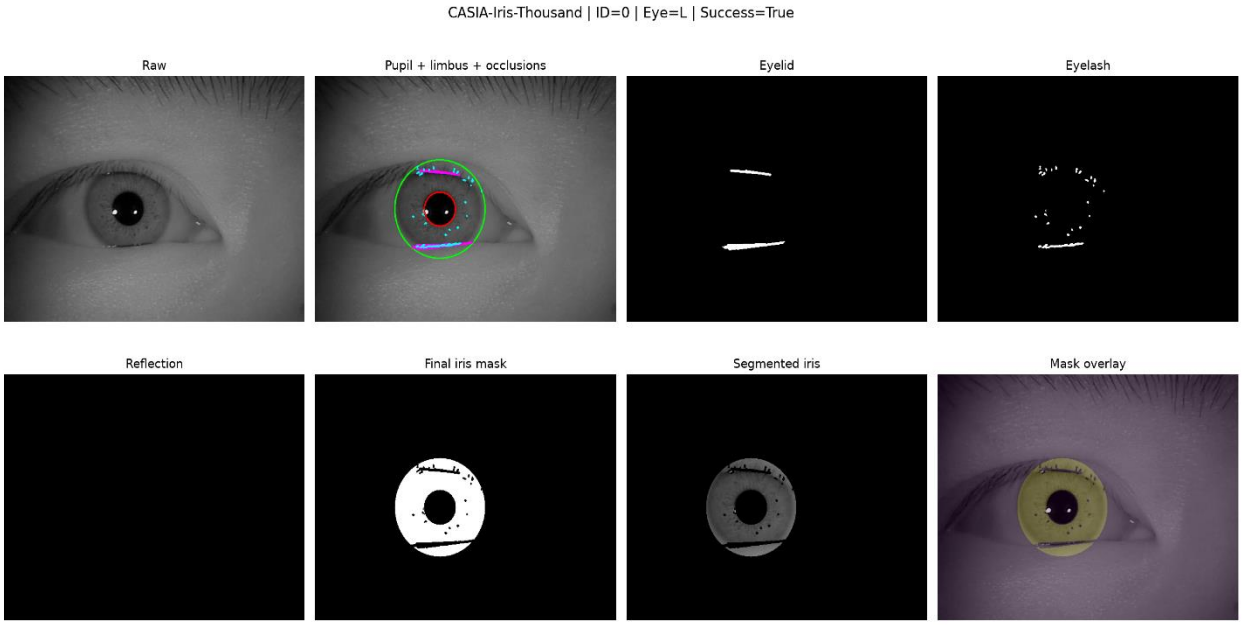


Figure 4: Sample of segmented iris

7.3.2 Iris Normalization – Notebook 03

Daugman's Rubber Sheet Model [1] was used to transform the segmented iris region into a fixed-size rectangular representation.

The polar coordinate mapping converts the circular iris region into a normalized image of 64 × 512 pixels. OpenCV remapping with bilinear interpolation was used for image intensities, while nearest-neighbour interpolation was used for masks.

Output: Normalized iris images, validity masks and normalization statistics.

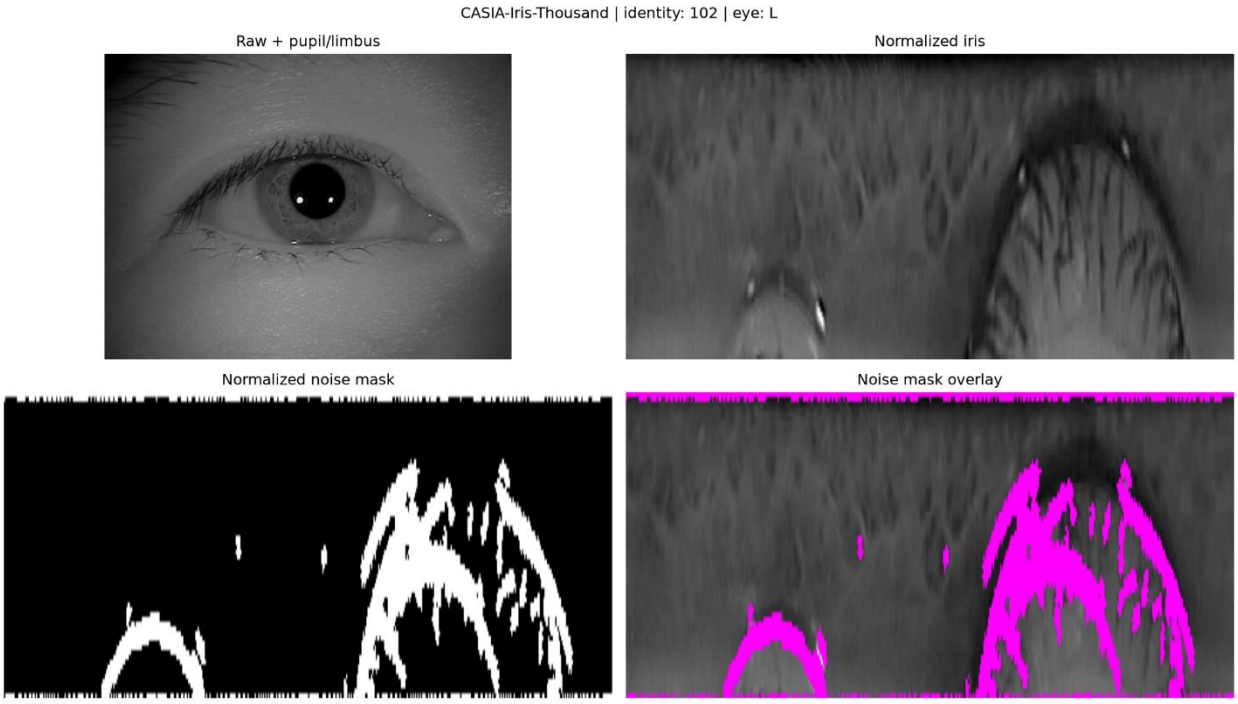


Figure 5: Sample of normalised iris

7.3.3 Image Enhancement – Notebook 04

Multiple enhancement techniques were implemented and compared to improve iris texture visibility.

- Histogram Equalization for global contrast enhancement.
- CLAHE with clip limit 2.0 and tile grid size 8 × 8.
- Gamma Correction using gamma = 1.2.
- Contrast Stretching using the 2nd and 98th percentiles.
- Denoising and sharpening operations.

The selected enhancement pipeline was applied to the normalized iris images.

Output: Enhanced iris images and comparison results.

7.3.4 Quality Assessment - Notebook 05

Image quality was evaluated using the following metrics:

- Blur: Variance of Laplacian.
- Focus: Tenengrad measure using Sobel gradients.
- Contrast: Standard deviation of valid pixel intensities.
- Illumination: Mean valid-pixel intensity.
- Occlusion: Fraction of invalid pixels.
- Reflection: Reflection area relative to the iris area.

These metrics were combined into a heuristic quality score ranging from 0 to 100. A configurable threshold of 50 was used to separate accepted and rejected images.

Output: Quality scores, accepted/rejected image records and quality distribution plots.

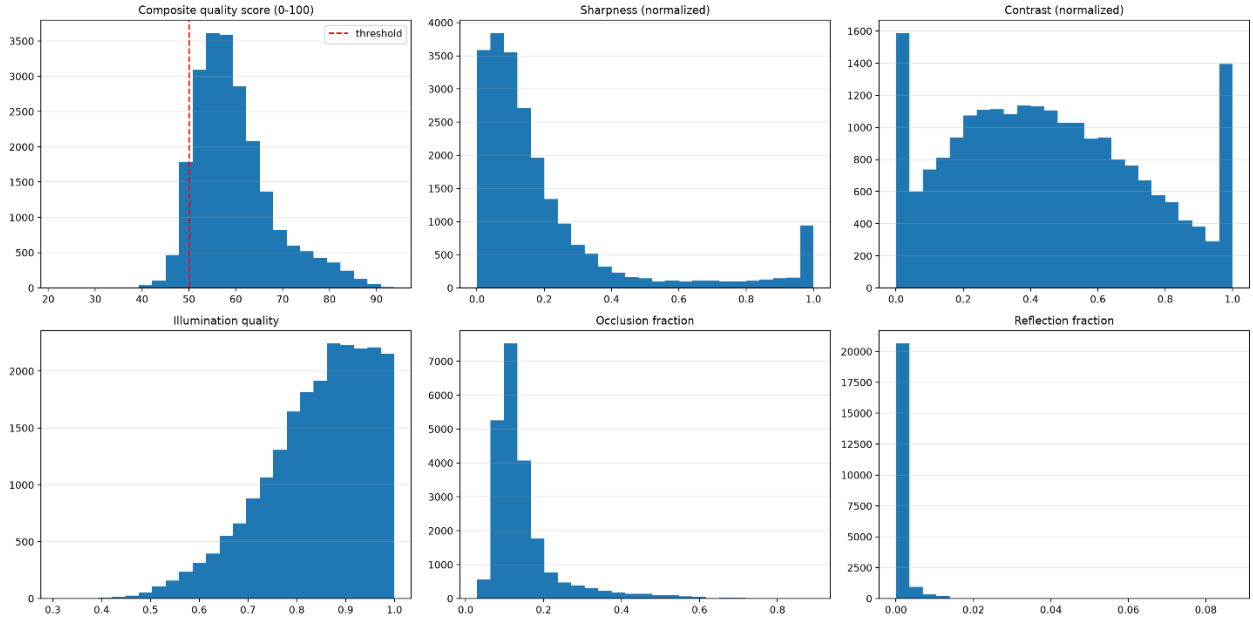


Figure 6: Quality distribution plots

7.4 Few-Shot Dataset Construction - Notebook 06

Quality-filtered images were organized into 1-shot, 2-shot, 3-shot and 5-shot configurations. Training, validation and testing partitions were created with checks to prevent data leakage.

Dataset manifests containing image paths, identity labels and partition information were generated for subsequent experiments.

Output: Few-shot datasets and corresponding manifests.

7.5 Baseline Recognition – Notebook 07

Baseline recognition models were trained using real iris images, with and without conventional data augmentation. Candidate architectures included ResNet50 [25], EfficientNet and Vision Transformer (ViT), with ArcFace [24] used as the principal identity-discriminative loss.

The models were evaluated using accuracy, ROC curves and Equal Error Rate (EER).

Output: Baseline model checkpoints and evaluation results.

7.6 Identity Representation Learning – Notebook 08

A β-Variational Autoencoder (β-VAE) framework was used to learn separate identity and appearance representations from normalized iris images [23].

The identity representation captures information intended to remain stable across images of the same iris, while the appearance representation captures visual variations.

The extracted latent vectors were used as conditioning information for synthetic image generation.

Output: Identity embeddings, appearance embeddings and trained model checkpoints.

7.7 Synthetic Iris Image Generation

7.7.1 Diffusion Model Training – Notebook 09

A conditional diffusion model was trained using real normalized iris images from the few-shot training partitions. The model learns to generate images through a progressive denoising process, using identity and appearance representations as conditioning information [17].

Output: Trained conditional diffusion model checkpoints.

7.7.2 Synthetic Image Generation – Notebook 10

Using the trained diffusion model, extra iris images were generated by maintaining identity conditioning but varying appearance related features.

These were saved as separate files along with their identity and generation manifests.

Output: Synthetic iris images and generation metadata.

7.7.3 Identity Verification – Notebook 11

The generated images are evaluated by calculating cosine similarities of their embeddings. Identity prototypes are built from the real images. The cosine similarity of each generated image and the corresponding identity prototype is compared to a threshold set using the real data.

Each generated sample is determined as either accepted or rejected.

Output: Accepted and rejected synthetic images, similarity scores and verification records.

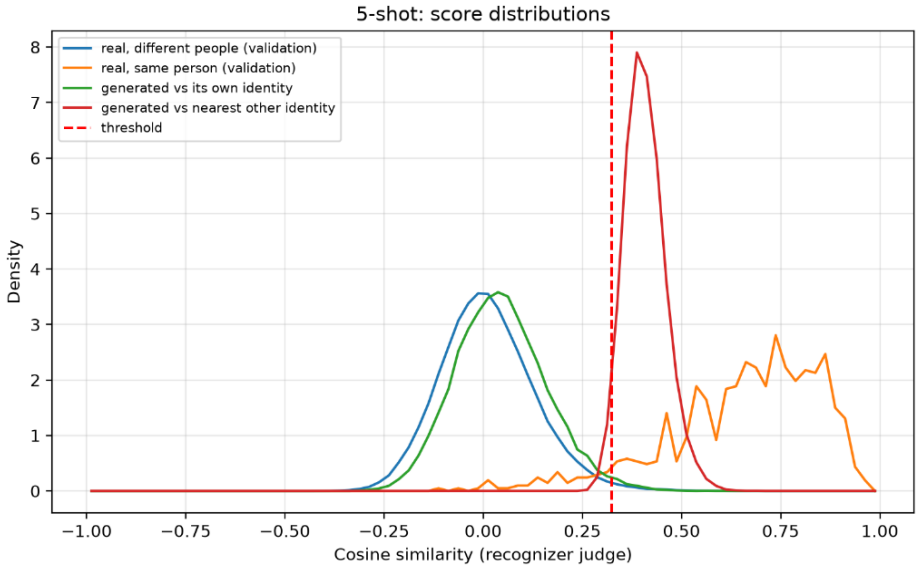


Figure 7: Cosine similarity for 5-shot

7.7.4 Synthetic Diversity Analysis – Notebook 12

The quality and diversity of generated images were examined using:

- FID (Fréchet Inception Distance) [19].
- KID (Kernel Inception Distance) [20].
- LPIPS (Learned Perceptual Image Patch Similarity) [21].
- MS-SSIM (Multi-Scale Structural Similarity) [22].
- Pixel-based image comparisons.

The analysis compared the generated samples and the identity-verified subset with real iris images.

Output: Diversity measurements and comparison visualizations.

7.8 Final Recognition Model – Notebook 13

The final recognition models were trained using real few-shot training images combined with accepted synthetic samples.

Candidate architectures included ViT, Swin Transformer, ConvNeXt and EfficientNetV2. The implementation also examined 256-dimensional feature embeddings and different loss configurations, including ArcFace, Triplet and Center losses.

Models were compared using validation performance, and the selected model was evaluated on held-out test data.

Output: Final recognition checkpoints and evaluation metrics.

7.9 Cross-Dataset Evaluation – Notebook 14

Cross-dataset evaluation was conducted by training recognition models on CASIA-Iris-Thousand and evaluating them on the unseen IIT Delhi dataset.

The evaluation included Rank-1 Accuracy, verification accuracy, EER and TAR@FAR. The experiments were used to examine generalization and the effect of dataset differences on recognition performance.

Output: Cross-dataset evaluation metrics and comparison plots.

7.10 Explainability and Feature Visualization – Notebook 15

Grad-CAM and Attention Rollout [26] were implemented to visualize important image regions and attention patterns. The learned feature embeddings were also visualized using t-SNE and UMAP [27] to examine identity clustering and intra-class variation.

Output: Heatmaps, attention visualizations and embedding plots.

7.11 Ablation Studies – Notebook 16

Controlled experiments were performed to investigate the contribution of different components of the framework. The experiments examined generator configurations, synthetic sample quantity, recognition backbones, loss functions and quality filtering.

Output: Ablation comparison tables and performance plots.

7.12 Performance Comparison – Notebook 17

Results from baseline, final recognition, cross-dataset and ablation experiments were consolidated into comparative tables and visualizations.

ROC curves, DET curves, confusion matrices, Precision-Recall curves, bar charts and box plots were generated. Results were exported in CSV, Markdown, LaTeX, PNG and PDF formats.

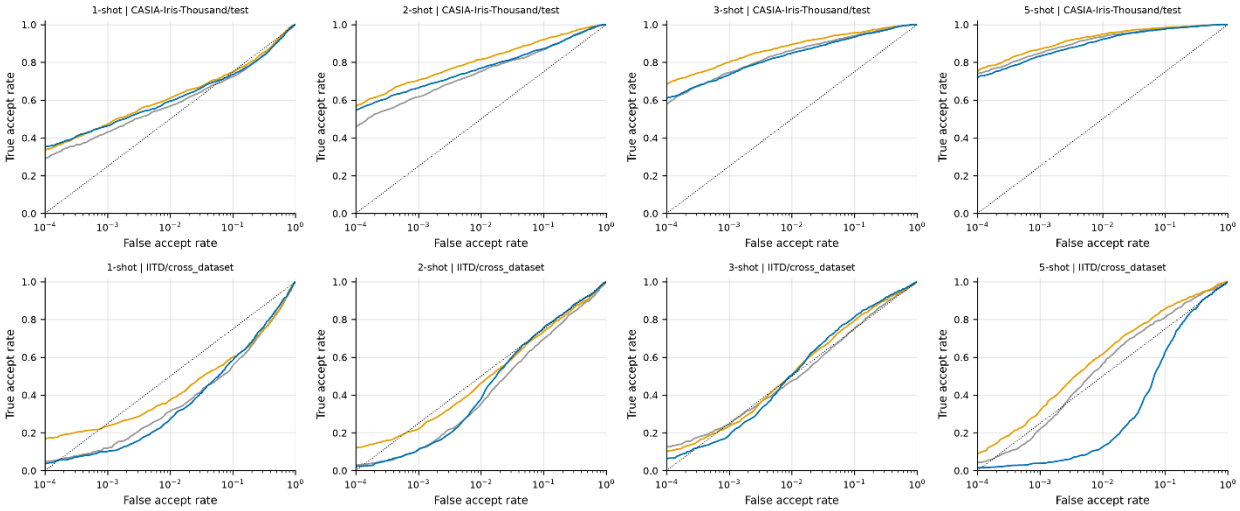


Figure 8: ROC Curves for CASIA-Iris Thousand and IITD

7.13 Statistical Analysis – Notebook 18

Statistical tests were performed to examine the reliability of experimental comparisons.

The implementation included:

- Paired t-test.
- Wilcoxon signed-rank test.
- Bootstrap confidence intervals.
- Cohen's d effect size.

These methods were used to analyze differences between experimental configurations.

7.14 Inference Demonstration – Notebook 19

An end-to-end pipeline was also implemented to demonstrate how iris recognition works using a new input image. This pipeline includes; pre-processing, feature extraction, identity matching,

prediction, confidence calibration, finding the nearest identities, unknown person detection and batch processing.

Output: Predicted identities, similarity scores, confidence information and batch prediction results.

7.15 Research Figure Generation – Notebook 20

The final notebook creates figures for publishing based on results from previous steps such as the pipeline overview, dataset stats, synthetic image comparisons, embeddings, ROC/DET curves, ablation studies, cross dataset analysis and failure cases.

Figures and captions are exported to a format ready to be included in a research report.

7.16 Overall Implementation Summary

The project has been split into 20 individual Jupyter notebooks which cover data exploration, preprocessing, creation of few shot data, baseline recognition, learning latent representations, creating synthetic images, verification and recognition, cross-dataset testing, explainability, ablation study, statistical analysis, inference demonstration and creation of figures for publication.

Table 9: Summary of Work performed in different notebooks

S. No.	Notebook	Work Performed
1	Dataset Exploration	Loaded and explored the CASIA-Iris-Thousand and IIT Delhi datasets. Analyzed image counts, subject distribution, image resolutions, left/right iris distribution, class imbalance and dataset characteristics.
2	Iris Segmentation	Implemented iris segmentation using pupil and limbus boundary detection. Applied eyelid, eyelash and reflection masking to extract the usable iris region and generated segmentation metadata.
3	Iris Normalization	Applied the Daugman Rubber Sheet Model to transform circular iris regions into fixed-size rectangular images (64 × 512). Generated corresponding noise masks and normalization statistics.
4	Image Enhancement	Implemented and compared CLAHE, histogram equalization, gamma correction, contrast stretching, denoising and sharpening techniques. Applied the selected enhancement pipeline to the normalized iris images.
5	Iris Quality Assessment	Evaluated image quality using blur, focus, contrast, illumination, occlusion and reflection metrics. Calculated composite quality scores and separated accepted and rejected images using a quality threshold.
6	Few-Shot Dataset Creation	Constructed 1-shot, 2-shot, 3-shot and 5-shot datasets using quality-filtered images. Created training, validation and testing splits with checks to prevent data leakage.
7	Baseline Recognition	Trained baseline iris recognition models using real images only, with and without conventional augmentation. Evaluated candidate backbones

		using ArcFace loss and established baseline performance metrics.
8	Identity Latent Space	Developed an identity and appearance representation learning model using a β-VAE architecture. Extracted identity and appearance latent vectors to support identity-conditioned synthetic image generation.
9	Diffusion Model Training	Trained a conditional diffusion model using identity embeddings and appearance codes. Saved trained generator models for producing synthetic iris images.
10	Synthetic Image Generation	Generated additional iris images by keeping identity representations fixed while varying appearance codes. Saved generated images along with metadata and visual comparisons.
11	Identity Verification	Evaluated generated images for identity consistency using recognition embeddings and cosine similarity with real identity prototypes. Classified synthetic samples as accepted or rejected.
12	Synthetic Diversity Analysis	Evaluated the quality, diversity and novelty of generated images using FID, KID, LPIPS, MS-SSIM and pixel-based comparisons. Compared all generated images with the accepted synthetic samples.
13	Final Recognition Model	Trained the final recognition models using real few-shot images combined with verified synthetic images. Evaluated different backbones and loss configurations and selected the best model using validation data.
14	Cross-Dataset Evaluation	Evaluated trained recognition models on the unseen IIT Delhi dataset to measure cross-dataset generalization. Analyzed Rank-1 accuracy, verification accuracy, EER and TAR at FAR.
15	Explainability	Applied Grad-CAM and Attention Rollout to examine important image regions. Used t-SNE and UMAP to visualize feature embeddings and analyze identity preservation and intra-class variation.
16	Ablation Studies	Conducted controlled experiments to investigate the effects of generator type, synthetic sample quantity, model backbone, loss functions and quality filtering on recognition performance.
17	Performance Comparison	Consolidated experimental results into comparative tables and figures. Generated ROC, DET and Precision-Recall curves, confusion matrices, performance charts and ablation visualizations.
18	Statistical Analysis	Evaluated the statistical reliability of model comparisons using paired t-tests, Wilcoxon signed-rank tests, bootstrap confidence intervals and Cohen's d effect size.
19	Inference Demonstration	Developed an inference pipeline that accepts iris images, performs preprocessing and predicts identities with confidence scores. Demonstrated identity matching and unknown-person detection.
20	Research Paper Figures	Generated publication-quality figures, diagrams and captions, including dataset statistics, synthetic image comparisons, embedding visualizations, evaluation curves, ablation charts and failure-case visualizations.

Results and Analysis

8.1 Introduction

This chapter presents the results obtained from the iris recognition and synthetic image generation pipeline. The experiments were conducted using the CASIA-Iris-Thousand and IIT Delhi iris datasets. The evaluation covers synthetic image generation, image quality and diversity, identity verification, and recognition performance under 1-shot, 2-shot, 3-shot, and 5-shot settings.

The main objective was to determine whether synthetic iris images could improve recognition when only a few real training samples were available. The results were compared with real-only training and conventional data augmentation. Cross-dataset evaluation was also performed to examine how well the models performed on data from a different dataset.

8.2 Dataset Preparation and Synthetic Image Generation

In order to understand the datasets, we have 20,000 images from CASIA-Iris-Thousand and 2,180 from IIT Delhi. This means in total we have 22,180 images in our dataset. We then preprocessed them by segmenting the iris, normalizing the data, improving the image quality and assessing the quality.

From the synthetic generation process we found that we had a total of 94,005 images generated for all four few-shot scenarios. As we can see from the training loss it has improved from around 0.876 to 0.0446. But just because the loss has improved doesn't mean that the generated images are realistic or useful for recognition.

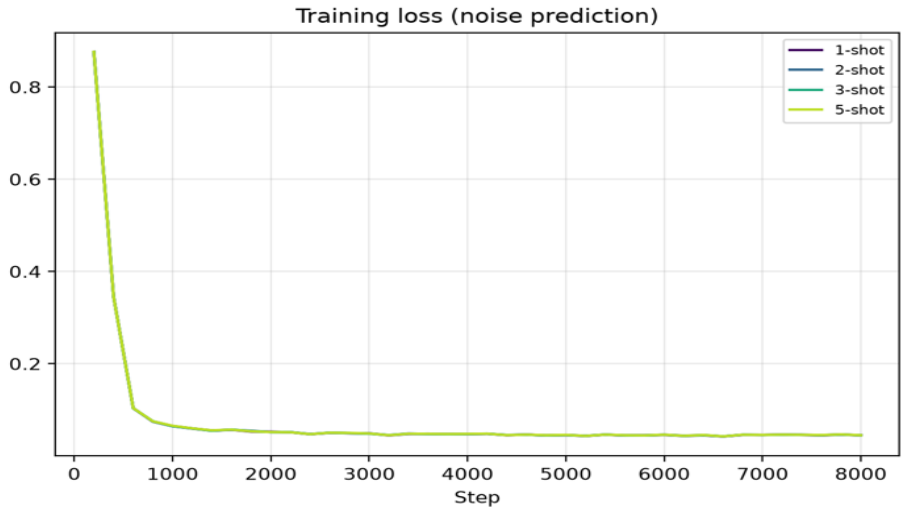


Figure 9: Training loss curve

The image statistics revealed differences between real and generated samples, particularly in sharpness. The average sharpness of generated images was approximately 3.04–3.70 times that of real images. This suggests that the generated images still differ noticeably from the original data.

The identity-verification stage accepted only 1,261 of the 94,005 generated images, giving an overall acceptance rate of approximately 1.34%. This low acceptance rate indicates that preserving identity-related characteristics remains a major challenge for the current generation pipeline.

8.3 Recognition Results on CASIA-Iris Thousand

The CASIA test results compare real-only training, conventional augmentation, and synthetic-data-assisted training.

Table 10: Recognition Performance on CASIA-Iris-Thousand

Setting	Training method	Rank-1 accuracy (%)	EER (%)
1-shot	Real only	46.36	20.75
1-shot	Conventional augmentation	50.98	19.56
1-shot	Real + synthetic	49.48	20.64
2-shot	Real only	65.27	11.96
2-shot	Conventional augmentation	73.34	8.59
2-shot	Real + synthetic	69.74	11.96
3-shot	Real only	78.00	7.07
3-shot	Conventional augmentation	83.90	5.60
3-shot	Real + synthetic	77.89	7.64
5-shot	Real only	89.43	3.65
5-shot	Conventional augmentation	89.49	2.93
5-shot	Real + synthetic	87.30	3.95

Overall, more real support images resulted in improvements in the recognition task. However, conventional augmentation resulted in improvements in Rank-1 accuracy compared to synthetic data assisted training (with lower EER values).

The Rank-1 accuracy results showed improvements from synthetic data assisted training compared to real training for both 1-shot and 2-shot scenarios. However, these

improvements were not consistent, and in the 3-shot and 5-shot scenarios the synthetic data assisted model had a slight drop in accuracy compared to the real training model.

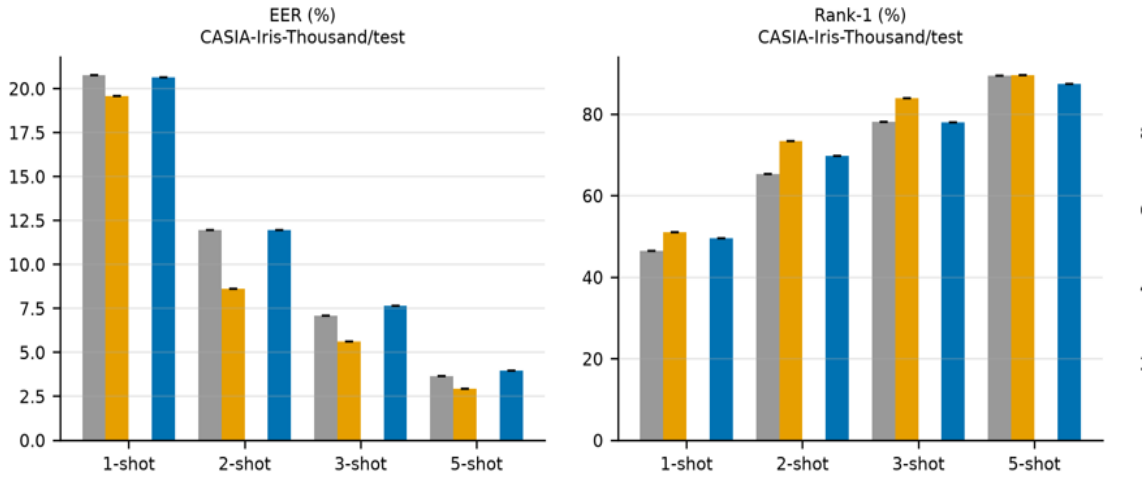


Figure 10: EER and Rank-1 Accuracy comparisons between Real only, Conventional Augmentation and Real+Synthetic images for CASIA-Iris Thousand

These results suggest that using the synthetically generated images did help to an extent in the low data regime however they did not perform as well as conventional augmentation on the CASIA test set.

8.4 Cross-Dataset Evaluation on IIT Delhi

Cross-dataset evaluation was performed to examine whether the models trained using CASIA data could generalize to the IIT Delhi dataset.

Table 11: Cross-Dataset Recognition Performance on IITD

Setting	Training method	Rank-1 accuracy (%)	EER (%)
1-shot	Real only	33.43	27.54
1-shot	Conventional augmentation	41.92	27.42
1-shot	Real + synthetic	36.60	26.14
2-shot	Real only	45.47	20.87
2-shot	Conventional augmentation	56.40	19.17
2-shot	Real + synthetic	53.13	18.02
3-shot	Real only	53.41	17.65
3-shot	Conventional augmentation	64.43	16.15

3-shot	Real + synthetic	59.94	14.75
5-shot	Real only	65.36	14.66
5-shot	Conventional augmentation	72.64	12.74
5-shot	Real + synthetic	55.00	20.45

Overall, when using cross-dataset testing, we can see that training with synthetic data gave the best performance in terms of having the lowest EER in all cases (1-shot, 2-shot, 3-shot). This suggests that training with synthetic data can be beneficial for verification.

However, traditional augmentation achieved the highest rank-1 accuracy across all scenarios. In the case of 5-shot testing the performance of the model trained with synthetic data was considerably worse with the rank-1 accuracy decreasing to 55.00% and an increase in EER to 20.45%.

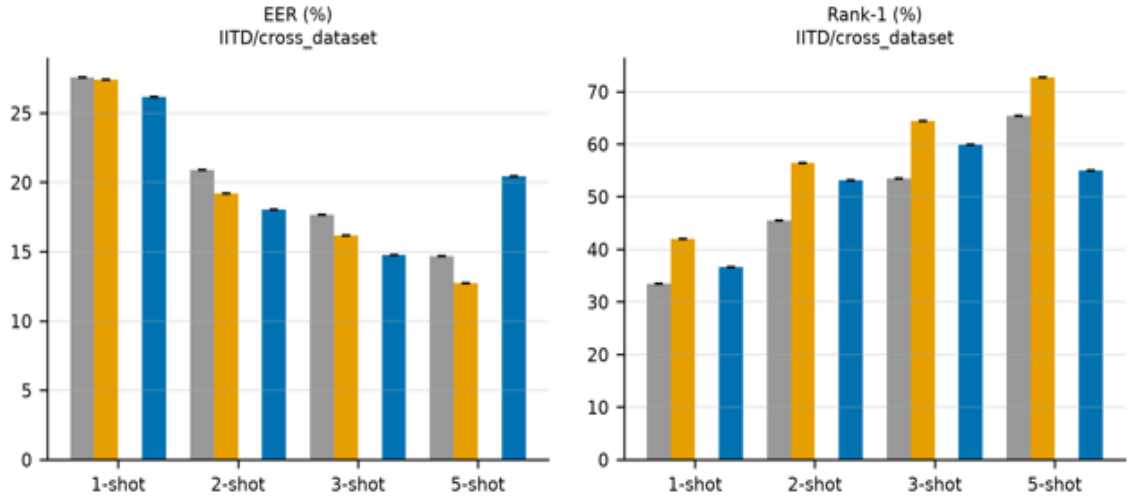


Figure 11: EER and Rank-1 Accuracy comparisons between Real only, Conventional Augmentation and Real+Synthetic images for IITD

Overall, it can be seen that training with synthetic data does not always give better generalisation to a new dataset. It depends on how many support images are used and what metric is being evaluated.

8.5 Image Quality, Diversity and Ablation Analysis

Analysis of image quality and diversity of the samples showed that the samples created were not perfect copies but were very different from real iris images. The FID scores varied from around 144.6 to 205.3 for the generated images compared to the real reference images

which only ranged from 19.7 to 33.6 within the Inception feature space. This shows there was a significant difference between the generated and real image distributions.

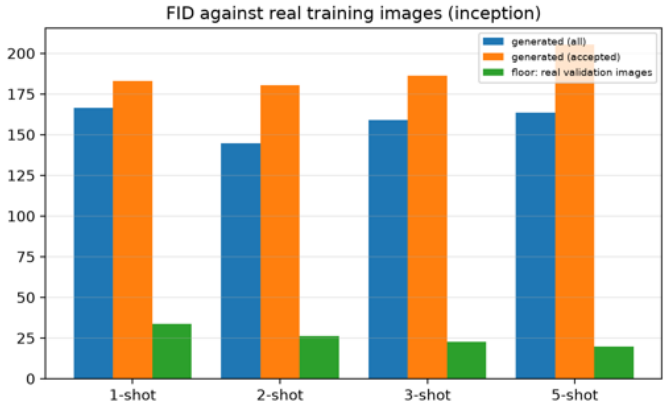


Figure 12: FID comparison

Analysis of diversity of samples showed that the identity separation scores were lower for the generated samples. While there was variation within the generated samples, there wasn't always good preservation of the identity within the samples. These scores need to be considered when looking at the recognition results.

The ablation study further supported these findings. While adding identity filtering helped certain recognition metrics, it didn't help consistently with both data sets. Conventional augmentation still performed better than synthetic assisted configurations. There was no clear optimum amount of synthetic images per identity found within the experiments as some configurations were restricted due to the small amount of accepted samples. Overall, it shows that simply having more samples isn't enough and the quality of the generated samples needs to be considered.

8.6 Discussion and Limitations

These experiments show that the pipeline can be used to create large amounts of synthetic iris images. However, results do not clearly outperform traditional data augmentation techniques. Whilst synthetic data did improve EER scores in some cross dataset scenarios, identification accuracy was not increased and results were poor in the 5-shot cross dataset scenario.

Limitations:

- Only one seed was used during training. This means that the model is not trained with random initialisation and there is no exploration of the variability in the training. A small amount of samples passed through the identity verification filter.

There are differences in image sharpness and sample distribution meaning the generated images do not closely match real iris images.

- The training sampler was not exactly the same for the baseline and final models being compared. This means we cannot be 100% sure that any differences in results are due to the synthetic data. More experiments would be needed to be run with the same training process and multiple independent runs to draw more accurate conclusions.

8.7 Summary of Experimental Findings

The results show that there is potential for synthetic iris images to be used to help tackle the small data issue in iris recognition however the current implementation does not provide any consistent improvements over traditional methods. The issue of image quality and distribution based on identity remains a concern.

Traditional augmentation provides the best and most consistent recognition results. While in some cases the addition of synthetic data helped improve EER in low shot cross dataset cases, this was not consistent. Future work would need to improve the image quality, increase the number of useful samples created and carry out controlled experiments using multiple training seeds to determine when synthetic data can be reliably used.

Conclusion and Future Scope

9.1 Conclusion

This project looked at the application of synthetic data to few-shot iris recognition. A pipeline was created that preprocessed iris data, performed baseline recognition, created synthetic data and finally evaluated the quality and diversity of the synthetic data. This was applied to both the CASIA-Iris-Thousand and IIT Delhi iris datasets.

The pipeline was able to generate 94,005 synthetic data samples. However, during testing it was found that generating more data did not necessarily improve performance. Only approximately 1.34% of samples were accepted during the identity verification step and there were clear differences in image quality between real data and synthetically generated data. This shows that maintaining useful data in regards to identities within the synthetic data is an issue.

The recognition results were varied. Using conventional data augmentation methods gave better results in terms of Rank-1 accuracies and lower Equal Error Rates (EERs) when testing on the CASIA dataset compared to synthetic data assisted training. However, during cross dataset testing on the IIT Delhi dataset, the synthetic data assisted training had lower EERs during 1-shot, 2-shot and 3-shot recognition. However, it still had worse Rank-1 accuracies. In addition, the performance dropped significantly during 5-shot recognition.

Overall, this suggests that synthetic data can be useful to provide extra training information however it is dependent on the dataset, metric used and how many supporting images are used. The quality of the synthetic data appears to be important.

This project gives an experimental insight into the potential and issues associated with generating synthetic iris data. While the results do not show any overall improvements over conventional data augmentation methods, there are some key areas that can be improved upon. Improving the preservation of identities, image quality and ensuring useful samples are generated. More controlled experiments need to be done to make further conclusions.

9.2 Future Scope

A few improvements can be made:

1) Quality of synthetic images: The synthetic images can be further improved by reducing the difference between the real and synthetic iris images. This includes reducing issues with image sharpness, texture, and distribution of features in the images. This could lead to creating better samples for training.

2) Identity preservation and selection: Currently, the identity verification process only accepts a small amount of the synthetic images. By improving how the samples are selected we can ensure that we are keeping the useful samples. We should also evaluate how we are selecting our samples to ensure we are not rejecting any valid samples.

3) Better recognition experiments: At the moment there is only one seed used for training in the comparisons and the training sampler is not exactly equal between the baseline and final model. In future experiments multiple random seeds should be used with identical training processes to get a better understanding of how much the synthetic data is contributing.

4) Cross dataset: The results for cross dataset show that just because we improve a certain evaluation method, it does not necessarily mean that we will see improvements elsewhere. By performing further experiments with varying training and testing datasets we can understand when it is most beneficial to train with the synthetic data.

5) Diversity and biometrics: We should look to understand if the synthetic images are providing meaningful diversity between identities whilst still maintaining realistic diversity within an identity. We should combine the image quality metrics with recognition results, identity separation and proper checks to see if we have memorised any of the training data.

6) Larger scale evaluation: Once we have improved image quality and identity preservation, we can perform evaluation using more identities, more datasets and more samples. This will let us understand if the benefits we are seeing are consistent when using more training data.

Overall, generating synthetic iris images is a promising area of research for overcoming limited training data in biometrics however, the usefulness of the synthetic images depends on the quality of the samples and how they affect the final recognition model. Future work should focus on ensuring we are preserving identities effectively and performing thorough evaluation.

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